

RF vs. Optical Downlinks in Very Low Earth Orbit: A Case Study on Link Budget, Atmospheric Losses, and Ground Infrastructure Requirements

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Abstract

Free-Space Optical (FSO) communication offers multi-gigabit throughput but remains vulnerable to atmospheric conditions, while X-Band RF provides robust all-weather connectivity at lower data rates. This paper presents a comparative link budget and contact analysis for both downlink architectures in Very Low Earth Orbit (VLEO, 180–400 km), evaluating achievable data volume over a one-year simulation against realistic ground segment configurations. Using a 2.5 Gbps optical terminal and a 200 Mbit/s X-Band transmitter, the optical link delivers 78.9 TB annually at a 400 km, 50° orbit, outperforming the RF link's 25.0 TB despite accumulating 3.7 times less contact time. The results confirm optical communication as the superior throughput solution for high-inclination VLEO missions, while RF remains indispensable for operational resilience and orbital flexibility.

Keywords

Satellite Communications — Radio Frequency — Optical Communications — Link Budget

1 Introduction

The growing data demands of modern Earth observation missions are straining the capacity of conventional RF downlinks. Spectrum congestion, regulatory constraints, and finite bandwidth availability increasingly limit achievable throughput for satellites in Very Low Earth Orbit (VLEO), an emerging operational regime typically defined as altitudes below 400 km. Free-Space Optical (FSO) communication addresses these limitations by operating outside the congested RF spectrum, offering narrow beam divergence and multi-gigabit data rates. However, optical links are highly sensitive to atmospheric conditions cloud cover, turbulence, and aerosol scattering can interrupt or degrade the link entirely whereas RF signals penetrate most weather with minimal loss. VLEO amplifies both the advantages and the challenges of each technology. The reduced slant range lowers free-space path loss for both links, but the optical signal must still traverse the full atmospheric column, and the high angular velocity of VLEO spacecraft imposes demanding pointing and acquisition requirements. This paper quantifies these trade-offs through a combined link budget and orbital contact simulation, comparing X-Band RF and 1064 nm optical downlinks across the full VLEO altitude and inclination trade space. The ground segments are based on the ESTRACK network for RF [1] and a combined EONN/GOGSN net-

work for optical communications. The primary metric is annual downlinked data volume, evaluated under realistic atmospheric and meteorological constraints.

2 Methods

The following section describes the hypothetical spacecraft configurations used for the VLEO link budget analysis. The radio frequency scenario is based on the European Space Agency ESTRACK network [1], while the optical case reflects currently available optical ground station networks. The study considers orbital altitudes between 180–400 km and inclinations from 0° to 100°. The analysis evaluates the annual downlinked data volume for both X-band and optical communication. Link feasibility is determined through the link budget, which defines achievable data rates under operational constraints and serves as the basis for estimating yearly data return. Due to the significant atmospheric drag in VLEO, continuous station-keeping is assumed. A propulsion system maintains constant orbital altitude throughout the mission, similar to GOCE [2] and SLATS [3]. The achievable data performance strongly depends on the selected ground station network. The networks considered here are chosen based on their availability for both RF and optical communication at the time of this study and are illustrated in the map below.

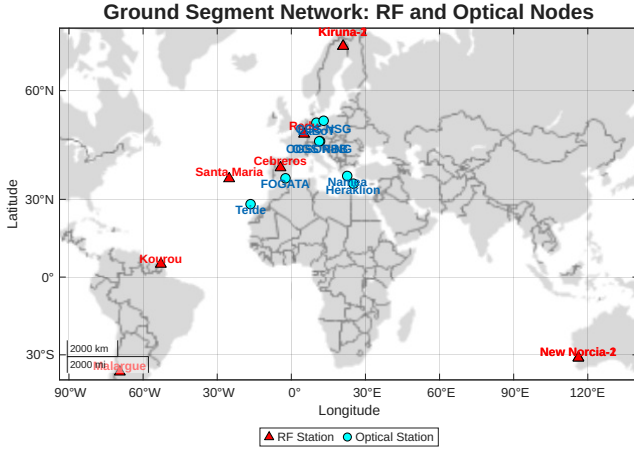


Figure 1. In red the radio frequency network, and in cyan the optical ground station network

2.1 RF Link Budget

The case study is based on earth observation mission on spacecraft within the cubesat and microsatellite class (up to 100 kg). This constraint also defines what potential configuration can be factored in. In the case of the ground segment, the selected ground station is chosen as the least performing station (lowest G/T -ratio, New-Norcia 2) within the ESTRACK network. This delivers a worst case scenario, which allows communication with the complete ESTRACK network used for the study.

2.1.1 Link Margin Calculation

The calculation is based on the International Telecommunication Union (ITU). This mainly focuses on Free Space Loss (FSPL) (ITU-R P.525-2 [4]) and the high data rates to be analyzed which increase the thermal noise temperature, decreasing the signal to noise ratio. In addition to these, environmental, attitude pointing, cable losses are being also considered. The calculation can be seen below:

$$\left(\frac{E_b}{N_0}\right)_{DL} = EIRP_{sat} - L_{FS} - L_{env} + \left(\frac{G}{T_{sys}}\right)_{gs} - k - R - L_p - L_{cables} \quad (1)$$

• Downlink Margin:

$$M_{DL}[dB] = \left(\frac{E_b}{N_0}\right)_{DL} - \left(\frac{E_b}{N_0}\right)_{req} \quad (2)$$

A market analysis was done to find an average performance value of a potential spacecraft, based on the communications specification which are mainly antenna

Table 1. Downlink Budget Parameters

Symbol	Description
$EIRP_{sat}$	Effective Isotropic Radiated Power
L_{FS}	Free Space Path Loss
L_{env}	Environmental Losses
$(G/T_{sys})_{gs}$	Ground Station Figure of Merit
k	Boltzmann Constant
R	Data Rate
L_p	Pointing Loss
L_{cable}	Cable Loss of Ground Station

gain, downlink power, data rate and lastly modulation scheme. These four parameters are the key contributors to the link and data budget. A set of commercial off-the-shelf antennas and transmitters from multiple vendors was analyzed to estimate representative performance values.

Table 2. Representative X-band transmitters for small satellites

Transmitter	TX Power (dBm)	Max Rate (Mbit/s)
IQ XLink-X [5]	30	200
NanoCom-X [6]	33	210
EnduroSat X-Band TX [7]	33	150
Satlab SRX-8 [8]	33	312
Safran N-XONOS [9]	33	500

Commercial CubeSat X-band transmitters typically provide output powers between 30–33 dBm and support data rates from 150 to 500 Mbit/s. Based on this survey, a transmitter power of **33 dBm** were assumed for the link budget analysis. Regarding the modulation scheme, **QPSK** with convolution code of 4, which was also result of the market analysis, was used. Based on a Bit Error Rate (BER) of 10^{-5} this results on required energy per bit ($\frac{E_b}{N_0}$) of **5 dB** [10]. As for the gain antenna of the spacecraft a value of **14 dBi** was assumed.

Table 3. Representative X-band spacecraft antennas for CubeSat missions

Antenna	Gain (dBi)
IQ X-Band Antenna [11]	18
GomSpace X-Band Patch [6]	13
EnduroSat X-Band Antenna [12]	12
Anywaves Compact X-Band [13]	16

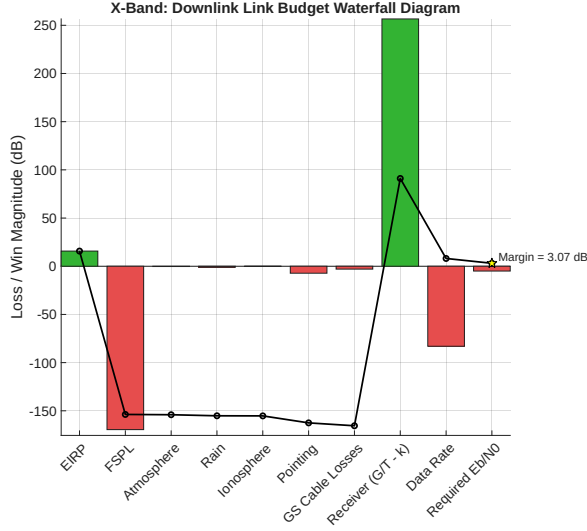


Figure 2. Downlink chain from spacecraft to ground station

Figure 3 represents the resulting link margins respective to the elevation of the New-Norcia 2 ground station, while also comparing different orbital heights. The result is an operational constraint that only allows downlink from an elevation of 20° . Figure 2 represents the worst case complete downlink chain, from spacecraft to ground station. The 3 key contributors to link budget are the gain over temperature of both spacecraft and ground station, which act as gaining factor, and the transmission rate and free space loss, which are seen as the contributor losses in comparison to the other losses. In order to comply to a link margin of 3 dB, the resulting maximum allowed downlink data rate is **200 Mbit/s**, which is also within the range of the market analysis.

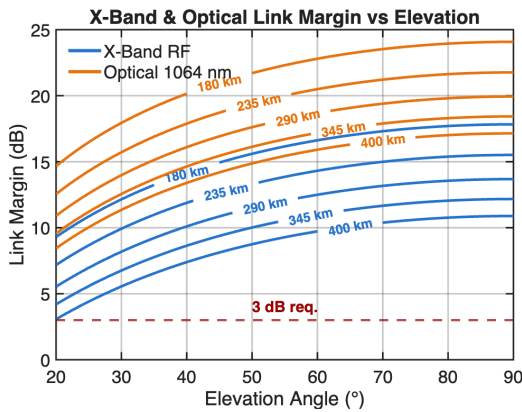


Figure 3. Link margin as a function of elevation angle for different orbital altitudes of X-Band and Optical Communication

2.2 Optical Link Budget

The optical communication model deviates from the RF approach by utilising a significantly higher carrier frequency (approximately 282 THz for a 1064 nm wavelength), which enables substantially higher directivity and throughput potential. The performance is evaluated in two stages: physical link feasibility via a link budget analysis, and temporal availability based on meteorological constraints at candidate ground stations.

2.2.1 Spacecraft Terminal and Link Parameters

To align with Space Development Agency (SDA) Optical Communications Terminal (OCT) standards applicable to VLEO/LEO orbits, the satellite is assumed to carry a laser terminal with a transmit power $P_t = 2.5$ W and a transmit aperture diameter of $D_t = 7$ cm [14]. The transmit gain G_t is given by:

$$G_t = \eta_a \left(\frac{\pi D_t}{\lambda} \right)^2 \quad (3)$$

where $\lambda = 1064$ nm is the operating wavelength and $\eta_a = 0.75$ is the aperture efficiency. The modulation scheme follows the SDA OCT standard, employing On-Off Keying Non-Return-to-Zero (OOK-NRZ) with direct detection.

Operating at 1064 nm, the terminal benefits from the reduced free-space path loss at VLEO altitudes (180-400 km) relative to higher orbital regimes. However, the optical signal must traverse the full atmospheric column, including the turbulent troposphere, making link performance highly sensitive to the elevation angle ϵ .

2.2.2 Ground Station Network and Optical Availability

The simulation utilises a distributed network of Optical Ground Stations (OGS) drawn from the European Optical Nucleus Network (EONN) and the German Optical Ground Station Network (GOGSN). Unlike RF links, which can operate through most cloud cover, optical downlinks require a Cloud-Free Line of Sight (CFLOS).

Mean optical availability for each site was calculated using the **CM SAF CLAAS 3.0** hourly cloud fraction dataset [15]. A site is considered optically viable when the Cloud Fraction Coverage (CFC) falls below a threshold of 40%. For stations where the aperture diameter was not specified in primary documentation, a standard value of 600 mm was adopted as representative of a typical EONN node configuration.

Table 4. Optical Ground Station (OGS) parameters and mean availability, computed using a CFC < 40% viability threshold. Apertures marked * are estimated default values.

Station	Aperture (mm)	Network	Avail.
Namea	600*	EONN	53.20%
Heraklion	600*	EONN	60.59%
Teide Obs.	600*	EONN	36.45%
FOGATA	600	GOGSN	55.31%
LaBoT	700	GOGSN	20.87%
OGS NBB	700	GOGSN	23.51%
OGS NSG	700	GOGSN	22.61%
OGSOP-NG	800	GOGSN	23.82%

2.2.3 Atmospheric Loss and Scintillation

The optical signal must traverse the full atmospheric column, with molecular absorption and Mie scattering from aerosols being most prevalent in the troposphere (below approximately 10 km altitude). The total atmospheric loss L_{atm} is modelled as a function of the zenith loss L_{zenith} and the slant-path airmass $M(\epsilon)$:

$$L_{atm}(\epsilon) [\text{dB}] = L_{zenith} \cdot M(\epsilon) \approx \frac{L_{zenith}}{\sin(\epsilon)} \quad (4)$$

where ϵ is the elevation angle. For a 1064 nm link under clear-sky conditions, L_{zenith} typically ranges from 0.3 to 0.6 dB depending on ground station altitude and local aerosol loading [16]. The design value adopted in this work is $L_{zenith} = 0.46$ dB, representative of a mid-latitude, clear-sky, near sea-level site. The approximation in (4) holds for elevation angles $\epsilon \geq 30$, which is enforced as the minimum operational elevation for the optical link in this study [17].

A fading margin $M_{fading} = 4$ dB is included to account for scintillation power fluctuations caused by atmospheric turbulence. This is particularly relevant in VLEO due to the high angular velocity of the spacecraft, which causes rapid traversal of turbulence cells along the slant path.

2.2.4 Link Margin Calculation

To establish a feasible communication link, the received power P_r must exceed the receiver sensitivity S_{req} for the target data rate. The free-space path loss is given by:

$$L_{FS} [\text{dB}] = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) \quad (5)$$

where d is the slant range and $\lambda = 1064$ nm. The received power is then:

$$P_r [\text{dBm}] = P_t + G_t + G_r - L_{FS} - L_{atm}(\epsilon) - L_{pt} - L_{sys} \quad (6)$$

where $L_{pt} = 4$ dB is the pointing loss and $L_{sys} = 5$ dB accounts for hardware and optical train losses. The receiver sensitivity threshold is set to $S_{req} = -20$ dBm, corresponding to a conservative bound for the 2.5 Gbps OOK-NRZ direct-detection receiver adopted in this study, derived from a 10 Gbps sensitivity baseline to ensure worst-case link margin estimation. The link margin is then:

$$LM [\text{dB}] = P_r - S_{req} - M_{fading} \quad (7)$$

A link is considered physically closed only when $LM \geq 3$ dB, as required by the SDA OCT Standard for compliant terminal designs [14].

Figure 3 presents the computed link margin as a function of elevation angle for both the optical and X-Band RF downlinks across the full VLEO altitude range (180-400 km) at a target data rate of 2.5 Gbps. All optical curves are valid for $\epsilon \geq 30$, consistent with the minimum operational elevation adopted for the optical link. Both links exceed the 3 dB closure requirement across all evaluated altitudes and elevation angles within their respective operational windows, confirming physical link feasibility throughout the VLEO regime.

2.3 Simulation

The total annual data volume V_{total} is the primary metric for comparing RF and Optical performance. In the simulation, the spacecraft's trajectory is propagated over one year. To ensure computational efficiency, the method avoids storing all state vectors of the propagation and instead evaluates communication opportunities during the propagation inside a step handler.

At each discrete time step, the satellite position is propagated and the elevation angle between the satellite and all ground stations in the network is computed. A ground station is considered eligible for communication when its elevation angle exceeds a predefined minimum threshold ϵ_{min} , representing the line-of-sight (LoS) constraint.

When this threshold is crossed, an Acquisition of Signal (AOS) event is detected and the corresponding start time is recorded for the specific ground stations. While the satellite remains visible, the algorithm continuously samples the elevation angle at each time step. These

values are accumulated into a figure of merit (FoM) for each ground station over the duration of the pass and weighted temporarily. This weighting method is adopted from [18]. In the referenced paper, at each time step t_n during a pass of total length m , the algorithm takes the elevation from each OGS to the satellite $\rho_{n,i}$ and multiplies it by a linearly decreasing weight w_{elev} , which starts at 1 for the first time step of the flyover and goes to 0 by the final time step.

$$g_i = \sum_{n=1}^m (\rho_{n,i} \cdot w_{elev}) = \sum_{n=1}^m \left(\rho_{n,i} \left(1 - \frac{n}{m} \right) \right) \quad (8)$$

This method however can only be applied if the total number of steps is known. To weight the elevation angles during the propagation, the summation is rearranged.

$$\sum_{n=1}^m \left(\rho_{n,i} \left(1 - \frac{n}{m} \right) \right) = \sum_{n=1}^m \rho_{n,i} - \frac{1}{M} \sum_{n=1}^m \rho_{n,i} n \quad (9)$$

In that way, the summation can continuously run until the elevation angle drops below ε_{min} . A Loss of Signal (LOS) event is then detected, marking the end of visibility for that ground station. At this point, the relevant sum is divided by the total number of steps. After that, the sums are combined to the FoM g_i for ground station i . The algorithm also evaluates all other ground stations that are currently in view and compares their accumulated FoMs. Additionally for OptoCom, each ground stations FoMs are adjusted for the optical availability. Based on the probability for clouds in the region of the ground stations, the FoMs can be multiplied by -1, indicating that for this specific pass, the ground station was not available for communication.

Then, the ground station with the highest FoM is selected as the optimal communication partner for that time period. In the case where the ground station that just had the LOS has the highest FoM, the corresponding contact interval—from its AOS to LOS—is accepted as a valid communication window for data downlink. If not, the pass is discarded which indicates that for that time period, a better downlink with another ground station could be achieved.

Since the satellite may simultaneously be visible to multiple ground stations, this procedure enables dynamic handover between stations. After a selection is made, all currently visible ground stations reset their AOS times to the current epoch, allowing a new evaluation cycle to begin without requiring storage of past trajectory data.

3 Results and Discussion

3.1 RF Downlink Performance and Data Volume

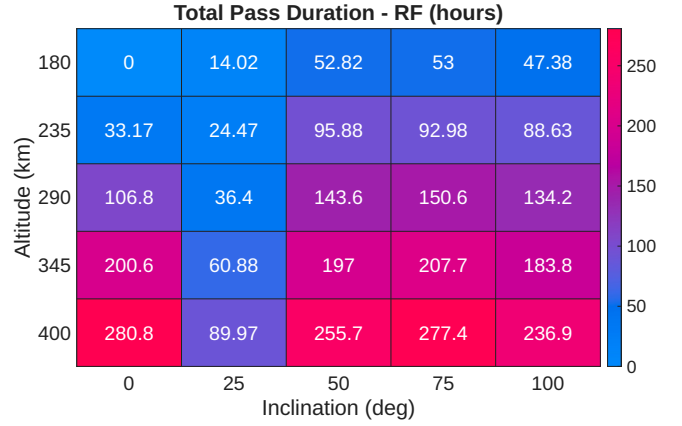


Figure 4. Sum of all passes for RF ground station network

3.1.1 Orbital Configuration and Ground Contact Analysis

Figure 4 presents the total annual RF ground contact duration across the evaluated VLEO altitude and inclination trade space. In contrast to the optical network, the ESTRACK ground stations provide non-zero contact at all evaluated inclinations, reflecting the all-weather capability of X-Band RF links and the wider beam divergence that relaxes pointing constraints.

A clear positive correlation is observed between both altitude and inclination and total contact time. The optimal configuration is a **400 km altitude, 75° inclination** orbit, yielding **277.4 hours** of annual contact through **4401** passes. This is approximately **3.7** times the contact time achieved by the optical network under its optimal configuration, driven primarily by the all-weather operability of the RF link and the wider spatial coverage of the ESTRACK network relative to the European OGS network.

3.2 Optical Downlink Performance and Data Volume

3.2.1 Orbital Configuration and Ground Contact Analysis

Figure 5 presents the total annual ground contact duration for optical downlinks across the evaluated VLEO altitude and inclination trade space. Contact time is expressed in hours accumulated over a 365-day simulation period. A clear trend is observed: higher altitudes and inclinations in the range of 40–60° consistently yield the greatest contact duration, driven by the larger

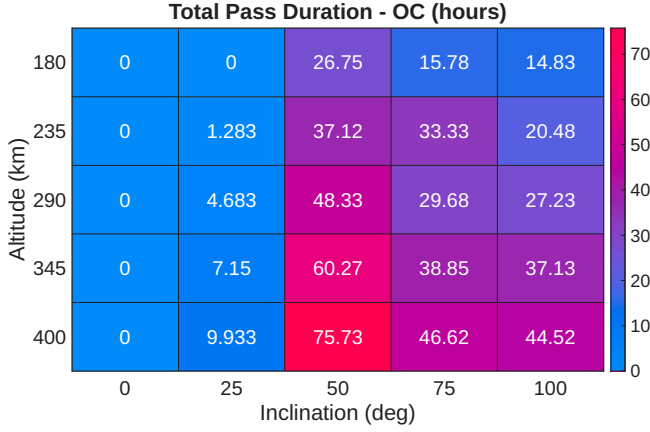


Figure 5. Sum of all passes for optical communication ground stations

ground track coverage and elevated horizon visibility at increased altitude.

The optimal configuration identified is a **400 km altitude, 50° inclination** orbit, yielding a total annual contact duration of **75.73 hours** across **2029 discrete passes**. At lower inclinations ($\leq 25^\circ$), contact with the primarily mid-latitude European OGS network drops to zero across all altitudes, reflecting the geometric incompatibility between low-inclination ground tracks and the northern European station locations. This underscores the sensitivity of optical network performance to the spatial alignment between the orbital plane and the ground station network geometry.

3.2.2 Effective Contact Time and PAT Overhead

Each optical pass requires an acquisition, pointing, and tracking (PAT) sequence prior to data transmission. A conservative PAT duration of 10 seconds per pass is assumed, consistent with demonstrated performance for modern fine-pointing terminals [19] which states a PAT duration of 8 Seconds with a ± 0.1 pointing error. For 2029 passes, the cumulative PAT overhead is:

$$T_{PAT} = N_{passes} \cdot t_{PAT} = 2029 \cdot 10 \text{ s} = 20,290 \text{ s} \approx 5.64 \text{ hours} \quad (10)$$

The effective data transfer time T_{eff} is therefore:

$$T_{eff} = T_{contact} - T_{PAT} = 75.73 - 5.64 = 70.09 \text{ hours} \quad (11)$$

representing a PAT overhead fraction of approximately 7.4%, which is modest given the large number of short passes characteristic of VLEO orbits.

3.3 Annual Data Volume

At the operational data rates of $R_{RF} = 200 \text{ Mbit/s}$ and $R_{OC} = 2.5 \text{ Gbps}$ respectively, the total annual data volumes are:

$$\begin{aligned} V_{RF} &= R_{RF} \cdot T_{contact} \\ &= 200 \times 10^6 \text{ bit/s} \times 277.4 \times 3600 \text{ s} \\ &\approx 199.7 \text{ Tbit} \approx 25.0 \text{ TB} \end{aligned} \quad (12)$$

$$\begin{aligned} V_{OC} &= R_{OC} \cdot T_{eff} \\ &= 2.5 \times 10^9 \text{ bit/s} \times 70.09 \times 3600 \text{ s} \\ &\approx 630.8 \text{ Tbit} \approx 78.9 \text{ TB} \end{aligned} \quad (13)$$

where $T_{eff} = 70.09$ hours accounts for the 5.64 hour cumulative PAT overhead across 2029 optical passes, while the full RF contact time of 277.4 hours is available for data transfer with no equivalent overhead.

It is important to note that the orbital configuration yielding the maximum downlinked data corresponds to a 400 km altitude and 0° inclination. This is due to the long-duration passes over the equatorial ground station in Kourou, which is consistently within the satellite's ground track. However, since this configuration relies on a single ground station and is therefore not considered realistic, it is excluded from further analysis.

3.4 Comparative Discussion: RF vs. Optical Downlinks

Table 5 summarises the key performance metrics for both downlink architectures under their respective optimal orbital configurations.

Table 5. Summary comparison of RF and optical downlink performance under optimal orbital configurations.

Parameter	X-Band RF	Optical
Optimal orbit	400 km, 75°	400 km, 50°
Annual contact	277.4 h	75.73 h
Effective transfer	277.4 h	70.09 h
Data rate	200 Mbit/s	2.5 Gbps
Annual data volume	$\approx 25.0 \text{ TB}$	$\approx 78.9 \text{ TB}$
Cloud dependent	No	Yes

Despite accumulating 3.7 times more annual contact time, the RF link delivers approximately 3.2 times less data than the optical link (25.0 TB vs. 78.9 TB). This inversion is driven by the 12.5 times data rate advantage of the optical link, which more than compensates for the

contact time lost to cloud cover and the narrower OGS network coverage.

The RF link nonetheless offers structural advantages not captured by the data volume metric alone. All-weather operability eliminates the stochastic availability loss that reduces optical contact to a cloud-fraction-weighted subset of geometrically visible passes, and the ESTRACK network provides non-zero contact at all evaluated inclinations, offering greater orbital flexibility than the European OGS network which becomes ineffective below 25° inclination.

The 78.9 TB optical figure should be interpreted as an upper bound, as solar interference, detector saturation, and background sky radiance have not been modelled and would each reduce effective throughput in practice. A more conservative estimate would narrow the throughput advantage over RF, though the optical link is expected to remain superior for high-inclination, data-intensive missions.

In summary, the two technologies are best viewed as complementary. RF provides robust, all-weather connectivity across all orbital geometries, while optical communication delivers order-of-magnitude throughput gains where atmospheric conditions and network geometry permit. For VLEO missions requiring both operational resilience and high data volume, a hybrid architecture combining an RF contingency link with an optical primary downlink represents the most capable solution.

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